

NEW HAVEN TWEED, CT, USA

WMO: 725045

Lat: 41.264N

Lon: 72.887W

Elev: 1

StdP: 101.31

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 14758

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.5	-9.9	-22.1	0.5	-10.7	-19.4	0.7	-7.9	11.1	-1.4	9.3	-0.2	4.0	320	0.464

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.4	30.9	23.5	29.4	22.7	28.2	22.0	25.0	28.8	24.2	27.5	23.5	26.6	3.4	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.8	18.6	27.0	23.0	17.7	26.0	22.4	17.1	25.6	75.6	28.9	72.5	27.7	69.8	26.5	27.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.8	7.7	6.9	-15.8	33.5	3.0	2.0	-18.0	34.9	-19.7	36.1	-21.4	37.2	-23.6	38.7		
			-16.6	26.1	2.8	0.8	-18.7	26.6	-20.3	27.1	-21.9	27.5	-24.0	28.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.5	-0.8	0.0	3.7	9.6	14.9	20.1	23.7	23.0	19.4	13.2	7.3	2.6
	DBStd	9.39	5.14	4.54	4.41	3.85	3.54	3.23	2.56	2.77	3.32	4.32	4.00	4.44
	HDD10.0	1219	337	279	198	52	3	0	0	0	0	18	99	233
	HDD18.3	2975	594	512	453	263	117	19	1	2	27	168	330	488
	CDD10.0	1750	0	0	4	40	156	302	424	403	282	117	19	3
	CDD18.3	464	0	0	0	1	12	71	167	147	59	8	0	0
	CDH23.3	3014	0	0	0	11	63	389	1227	1021	287	16	0	0
	CDH26.7	684	0	0	0	1	8	78	330	226	39	2	0	0
Wind	WSAvg	3.1	3.4	3.4	3.6	3.4	2.9	2.6	2.5	2.6	2.9	3.2	3.3	3.4
Precipitation	PrecAvg	1065	74	79	89	98	99	88	88	80	86	90	101	96
	PrecMax	1457	155	135	234	272	268	322	222	199	196	211	220	196
	PrecMin	703	6	14	18	36	20	6	21	18	23	7	9	26
	PrecStd	212	40	28	46	52	62	72	54	46	52	52	59	51
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.6	13.2	17.7	24.7	27.6	31.0	33.4	32.0	29.8	25.3	19.0	15.5
		MCWB	10.1	9.3	12.2	15.5	19.2	22.8	24.3	24.3	23.6	20.1	16.0	12.7
	2%	DB	10.2	10.4	14.2	20.7	25.1	28.9	31.3	30.2	27.7	22.9	17.2	13.0
		MCWB	8.6	7.7	9.4	13.4	18.6	21.3	23.6	23.5	21.6	20.0	14.4	11.2
	5%	DB	8.1	8.4	12.1	18.0	22.9	27.1	29.8	29.0	26.2	21.6	15.8	11.4
		MCWB	6.0	5.7	8.2	11.9	17.1	20.6	22.9	22.7	20.9	18.6	13.4	9.8
	10%	DB	6.6	6.6	10.2	15.9	21.1	25.6	28.4	27.8	24.9	20.0	14.3	9.8
		MCWB	4.6	3.9	6.7	10.7	15.8	19.7	22.3	22.2	20.4	16.8	11.8	7.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.5	10.9	13.7	17.3	21.3	24.0	26.0	26.1	24.7	22.1	17.3	13.8
		MCDB	12.0	11.7	16.3	22.7	25.9	29.3	31.5	30.0	28.1	23.4	18.8	14.5
	2%	WB	8.9	8.1	10.8	15.0	19.8	22.8	24.9	25.1	23.6	20.7	15.4	11.7
		MCDB	9.8	9.9	12.9	18.4	23.6	27.0	29.5	28.5	26.1	22.7	16.7	12.8
	5%	WB	6.6	6.2	8.9	13.4	18.4	21.9	24.1	24.3	22.6	19.2	13.9	10.2
		MCDB	7.9	8.0	11.1	16.5	21.8	25.5	27.8	27.3	24.8	21.1	15.4	11.4
	10%	WB	4.8	4.3	7.5	11.8	17.0	20.9	23.5	23.6	21.6	17.6	12.4	7.9
		MCDB	6.5	6.3	9.6	14.4	19.8	23.9	26.9	26.3	23.8	19.6	14.0	9.4
Mean Daily Temperature Range	5% DB	MDBR	7.6	8.2	8.3	9.2	8.9	8.5	8.4	8.4	9.0	8.8	8.6	7.3
		MCDBR	9.2	10.3	11.0	12.9	11.7	10.2	9.7	9.1	9.7	9.1	9.6	8.6
		MCWBR	8.4	8.1	7.5	7.0	6.1	5.0	4.3	4.4	5.6	6.4	7.9	8.3
	5% WB	MCDBR	9.2	10.0	9.7	10.6	10.3	8.9	8.3	7.5	7.4	7.6	8.2	8.2
		MCWBR	8.9	8.7	7.8	7.0	6.1	4.8	4.1	4.0	4.9	6.1	7.8	8.6
Clear-Sky Solar Irradiance	taub	0.309	0.318	0.341	0.377	0.427	0.471	0.489	0.462	0.390	0.349	0.319	0.307	
	taud	2.474	2.454	2.440	2.371	2.262	2.176	2.138	2.204	2.378	2.508	2.544	2.539	
	Ebn at Noon	848	895	909	894	853	812	793	803	845	848	832	823	
	Edh at Noon	76	90	102	117	134	146	151	137	108	85	71	66	
All-Sky Solar Radiation	RadAvg	1.85	2.72	3.72	4.81	5.37	5.85	5.97	5.31	4.36	2.95	2.05	1.55	
	RadStd	0.15	0.25	0.29	0.51	0.53	0.57	0.32	0.32	0.35	0.28	0.17	0.14	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.96	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+137	N/A
(47)	Regional (71 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+107	+58

Nomenclature: See separate page